

PRELIM EXAM

CS Elective 2: Mobile Application Development
CS 4151 | 4-335 | 6:00 - 8:30 PM (Tue. & Thu)

E-COMMERCE APPLICATION: SHOP BROWSER

INSTRUCTIONS:

Build an e-commerce shop application centering on product browsing experience. You have the freedom to choose the main product of the said e-commerce application. This project demonstrates all the topics we have covered so far, namely Basic Widgets, Navigation 1.0 and 2.0, Stateless vs. Stateful Widgets, Design Theming, and Responsive and Adaptive Design. The user flow must be:

Home (product grid) → Product detail → Add to Cart → Cart → Checkout Confirmation

Screen	Required functionalities
Home	• Displays all products in a grid • Use GridView (or GridView.builder) for the product list • 2-column grid for phones and 3+ columns for tablet • Clicking a product navigates to Product Detail Page
Product Detail	• Shows required information for each product such as: • Image • Price • Other details (optional) • Reached via Navigation 2.0 (go_router package)
Cart	• Shows items added, with quantities and subtotal • Stateful quantity control (+ or -) • Running total updates live
Checkout Confirmation	• Final screen after checkout • Displays simple summary of: ○ Cart items ○ Subtotal per item ○ Total

	○ Confirmation message ● Reachable only after at least one item is in the cart
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- Use a Scaffold with an AppBar on every screen.
- Use Card, Image, and Text widgets to present product info cleanly.
- Anything that never changes after being built (e.g. a single product card's static display) should be a StatelessWidget.
- Anything that changes based on user interaction (cart quantity, add-to-cart button state, theme toggle) should be a StatefulWidget, with clear justification you can explain if asked.
- Implement Navigation 2.0 throughout the project.
- Define a single ThemeData (colors, text styles, button styles) applied at the MaterialApp level — no hardcoded colors scattered through individual widgets.
- Include a working light/dark mode toggle, accessible from the home screen (via AppBar), that changes the whole app's appearance immediately.
- The Home screen's product grid must show 2 columns on phone-width screens and at least 3 columns on tablet-width screens.
- Use LayoutBuilder or MediaQuery to detect width and adjust the column count.

SUBMISSION & PRESENTATION:

- Create a new branch to your existing GitHub repository for this subject.
- Name the new branch as PrelimExam
- Submit GitHub repo link in this bin.
- Half of the project (incl. Design theme, home screen, routing to an empty details page) should be completed and presented by September 3, 2026 (Thursday) in class.
- The rest of the project must be completed by September 8, 2026, 11:59 PM.

RUBRICS:

<i>Rubric Item</i>	<i>Excellent</i>	<i>Developing</i>	<i>Needs improvement</i>	<i>Total</i>
<i>Design Theming</i>	20 pts.	12 pts.	6 pts.	20 pts.
<i>Stateless vs Stateful Widget</i>	10 pts.	7pts.	4 pts.	10 pts.
<i>Routing</i>	15 pts.	8 pts.	5 pts.	20 pts.
<i>Screens and corresponding functionalities</i>	25 pts.	15 pts.	7 pts.	25 pts.
<i>Proper widget</i>	15 pts.	8 pts.	5 pts.	15 pts.

<i>use (e.g. use of Scaffold, Card, MediaQuery, LayoutBuilder, GridView.builder, etc.)</i>				
<i>Responsive design (mobile to tablet sizes)</i>	10 pts.	7pts.	4 pts.	10 pts.
			<i>TOTAL</i>	<i><u>100</u></i>